SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING, KALAVAKKAM
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
III Year CSE – A, B, C Sections (VI Semester)

Academic Year 2025-26

UCS2611- INTERNET PROGRAMMING LAB

Project Title: Circle

IP Mini Project: Complete Project Documentation

Date: 04/04/26

Team Members

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1. Problem Statement

Modern social media platforms are largely driven by engagement metrics such as likes, shares, and algorithmic recommendations. While these features increase user activity, they often compromise authenticity and overwhelm users with irrelevant content. Additionally, privacy concerns arise due to public sharing and weak access control mechanisms. There is a clear need for a platform that prioritizes real relationships, minimizes distractions, and ensures strict privacy.

2. Executive Summary

Circle is a friends-first social networking application designed to promote meaningful and private interactions among users. Unlike traditional platforms that rely heavily on engagement-driven algorithms, Circle focuses on authenticity by prioritizing connections between real friends. The platform provides a chronological feed, ensuring transparency and eliminating manipulation through recommendation systems. By enforcing privacy at the database level, Circle guarantees that user data is only accessible to authorized connections, creating a safe and controlled environment for communication.

3. Objectives

- Provide a private-by-default user experience where only accepted friends can interact.

- Eliminate algorithmic feeds and replace them with a chronological timeline.

- Implement strong database-level security using Row Level Security (RLS).

- Enable seamless sharing of content including text and media.

4. Product Scope

The MVP includes authentication, profile management, friend connections, posting, and engagement features such as likes and comments. A visual graph representation of connections is also included. Advanced features such as recommendations, advertisements, and analytics are excluded.

5. Target Users

Students and small communities who prefer a private and distraction-free social networking experience.

6.Tech Stack

Technology	Purpose

React + TypeScript	Frontend development
Tailwind CSS	Styling
Zustand	State management
Zod	Validation
D3	Graph visualization
Supabase	Backend services

7. System Architecture

The system follows a modular architecture consisting of UI components, hooks, service layers, and backend services. The UI handles rendering and user interaction, hooks manage state and orchestration, and services interact with Supabase for database and storage operations.

Row Level Security (RLS) is enforced at the database layer, ensuring that only authorized users can access specific data. This architecture ensures scalability, maintainability, and strong security.

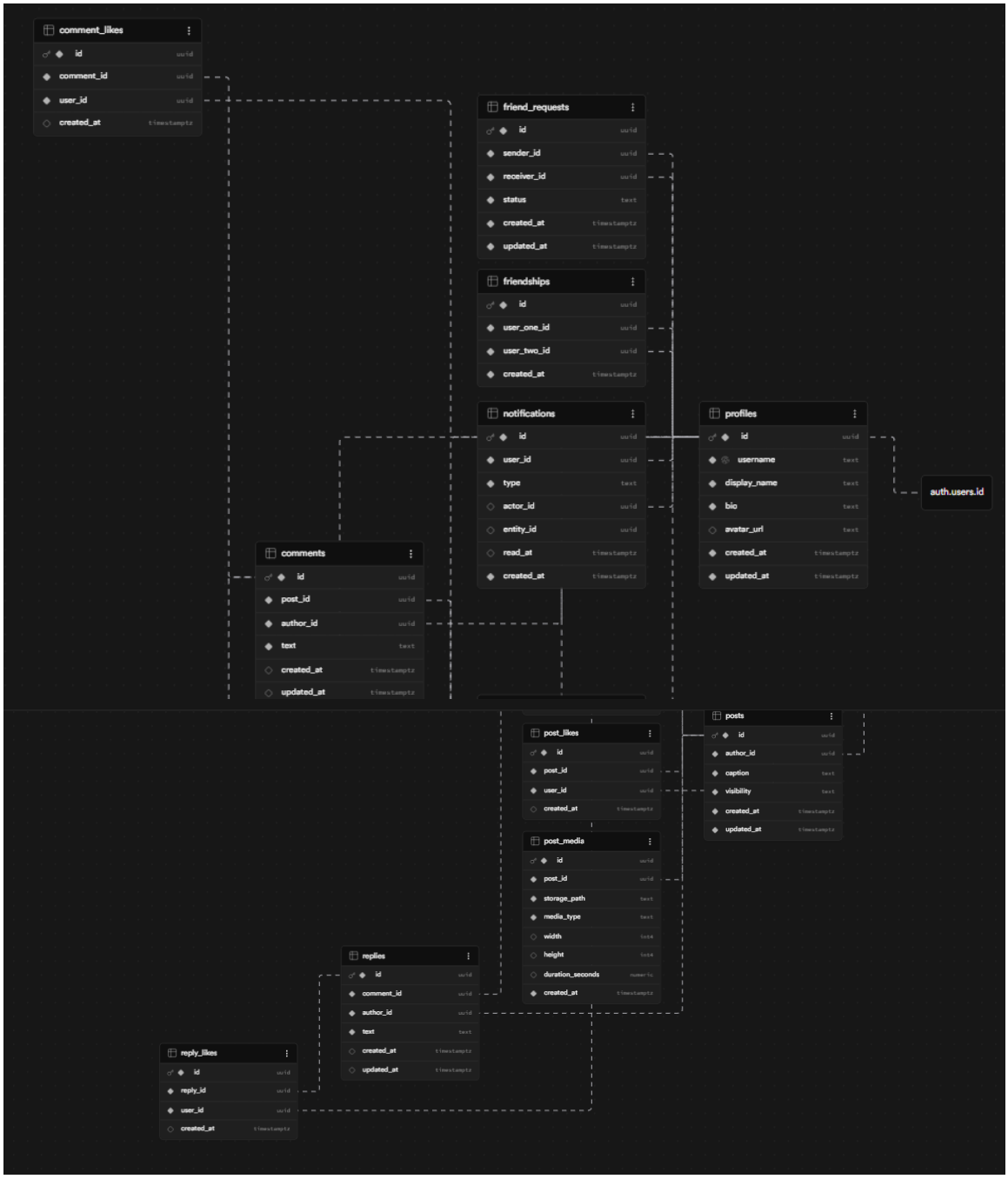
8. Database Design & Security

The database is structured using PostgreSQL via Supabase. Core tables include profiles, friendships, posts, and media, while engagement tables handle likes, comments, and replies.

RLS policies enforce strict access control, ensuring users can only read/write their own data or their friends' data. This prevents unauthorized access and ensures privacy.

Triggers are used to automatically update timestamps, and storage policies control media uploads and access.

ER Diagram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reply_likes
id uuid
reply_id uuid
user_id uuid
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9. Features Implemented

Authentication

Authentication is handled using Supabase Auth with email and password login. The system manages sessions, prevents duplicate submissions, and handles authentication state changes using React hooks.

Profile Management

Users can create and update profiles including username, display name, bio, and avatar. Avatar uploads are handled using Supabase Storage with secure access policies.

Friends System

Users can search for other users, send friend requests, and accept or reject requests. Accepted friendships are stored in a dedicated table with constraints ensuring uniqueness.

Feed System

The feed displays posts from the user and their friends in chronological order. It fetches posts, media, and engagement data efficiently using service layers.

Posting & Media

Users can create posts with optional image or video uploads. Media is stored in Supabase Storage and linked via database records.

Engagement

Users can like posts, comment, and reply. Engagement data is stored in dedicated tables with optimized queries and optimistic UI updates.

Graph Visualization

A force-directed graph built using D3 visualizes the user's network, including direct and extended connections. It supports zoom, drag, and interactive highlighting.

10.Planned Features

Future features include

- Notifications
- enhanced RLS policies
- media compression
- backend edge functions for optimized queries.
- A private chat between users
- User stories

11.Setup & Run

1. Create a Supabase project.

2. Run SQL migrations for schema and RLS.

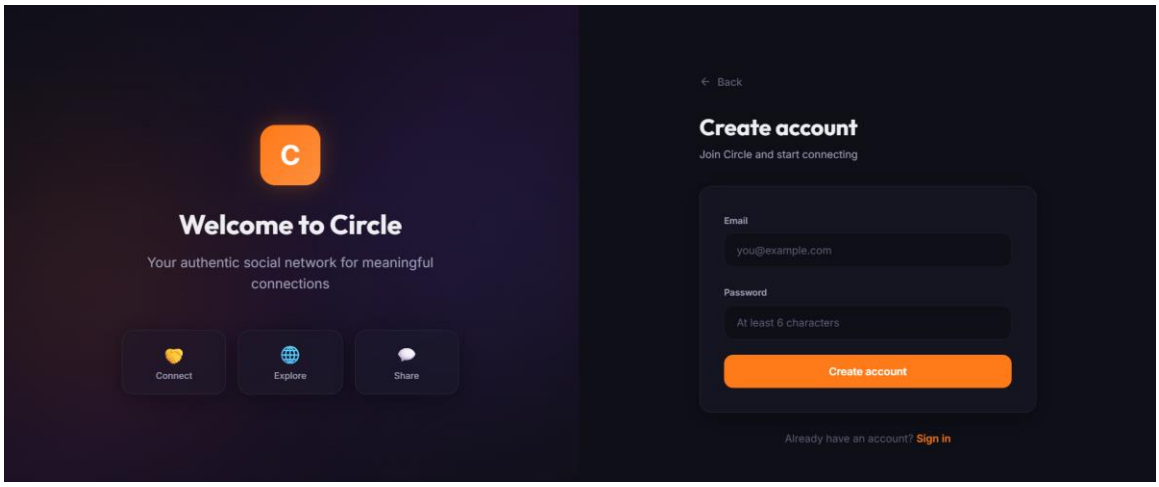
3. Configure environment variables.

4. Install dependencies and run the frontend using the command “npm run dev”.

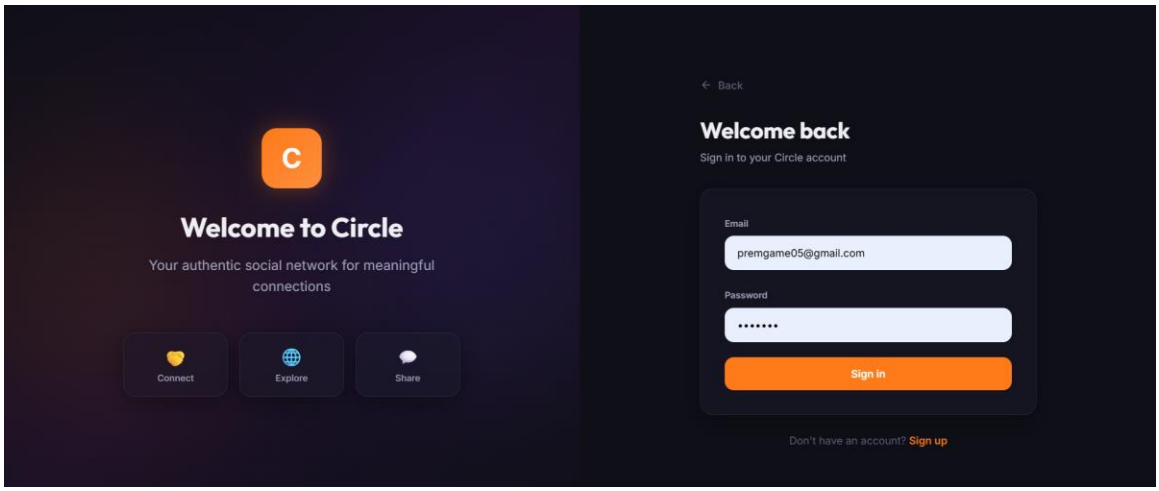
12. Screenshots

Entry page

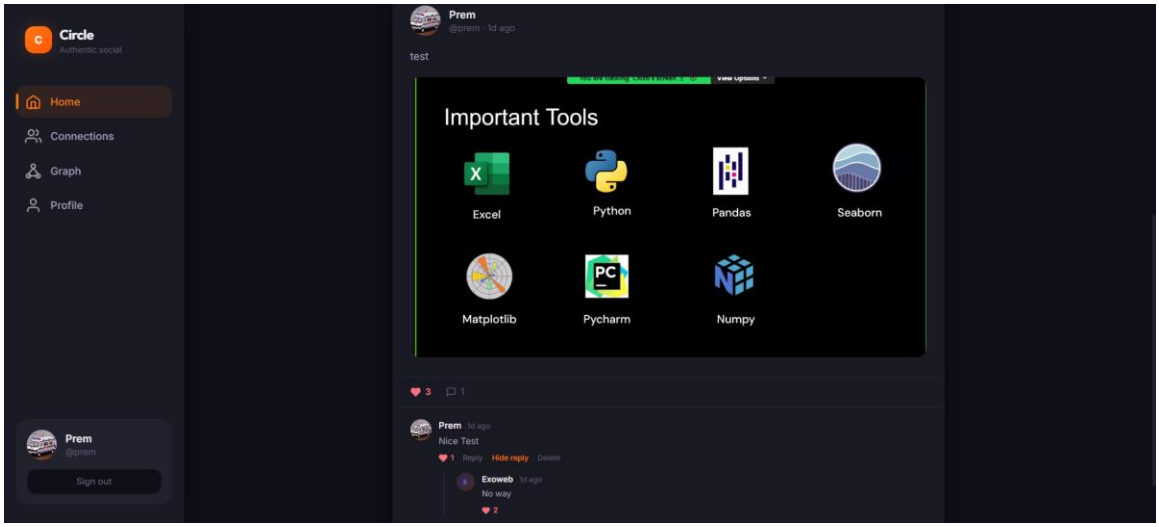
Create Account



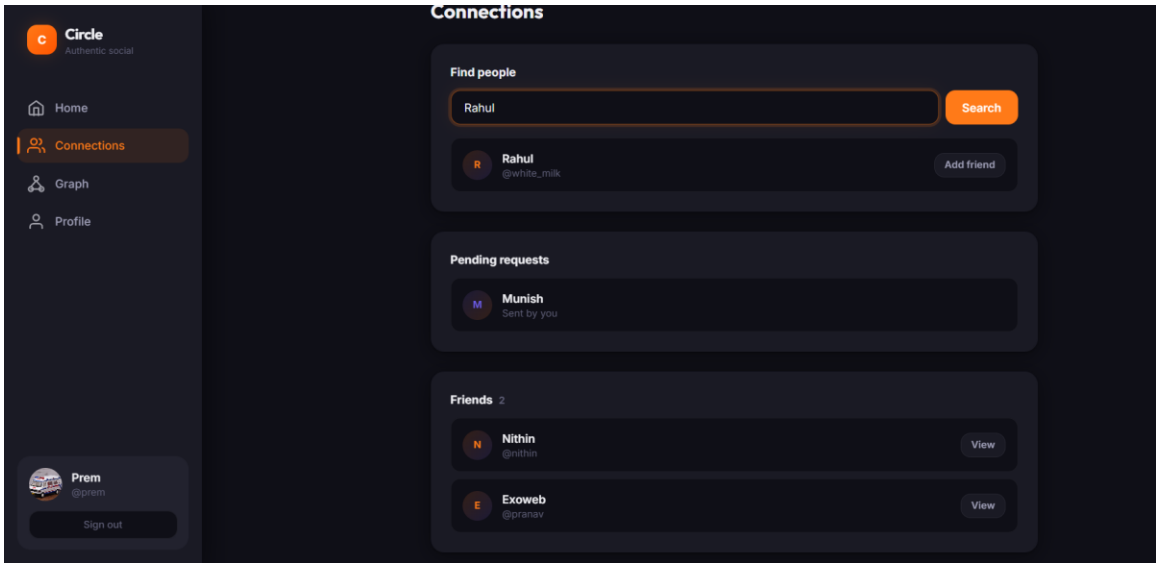
Sign in



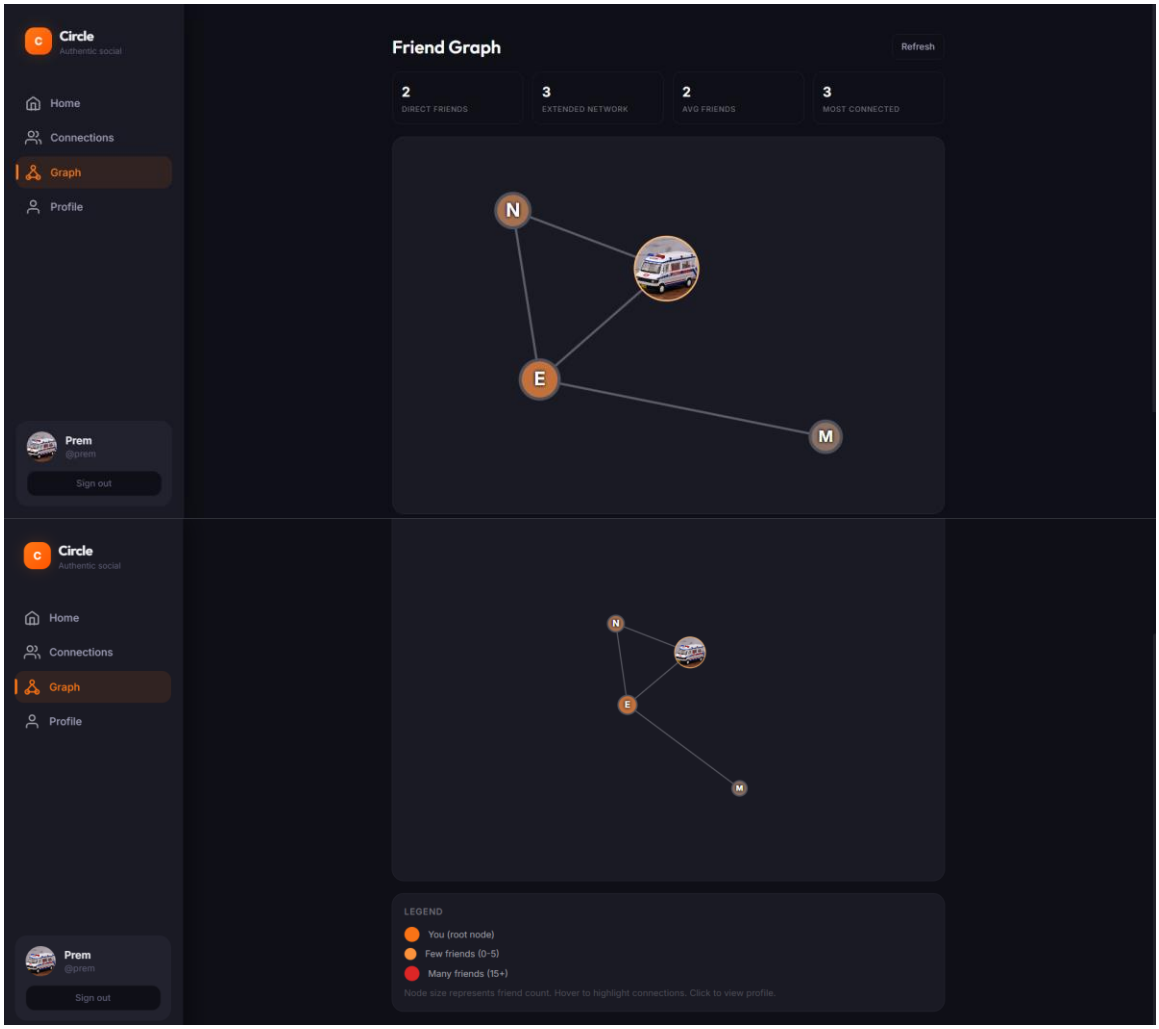
Home page with feeds



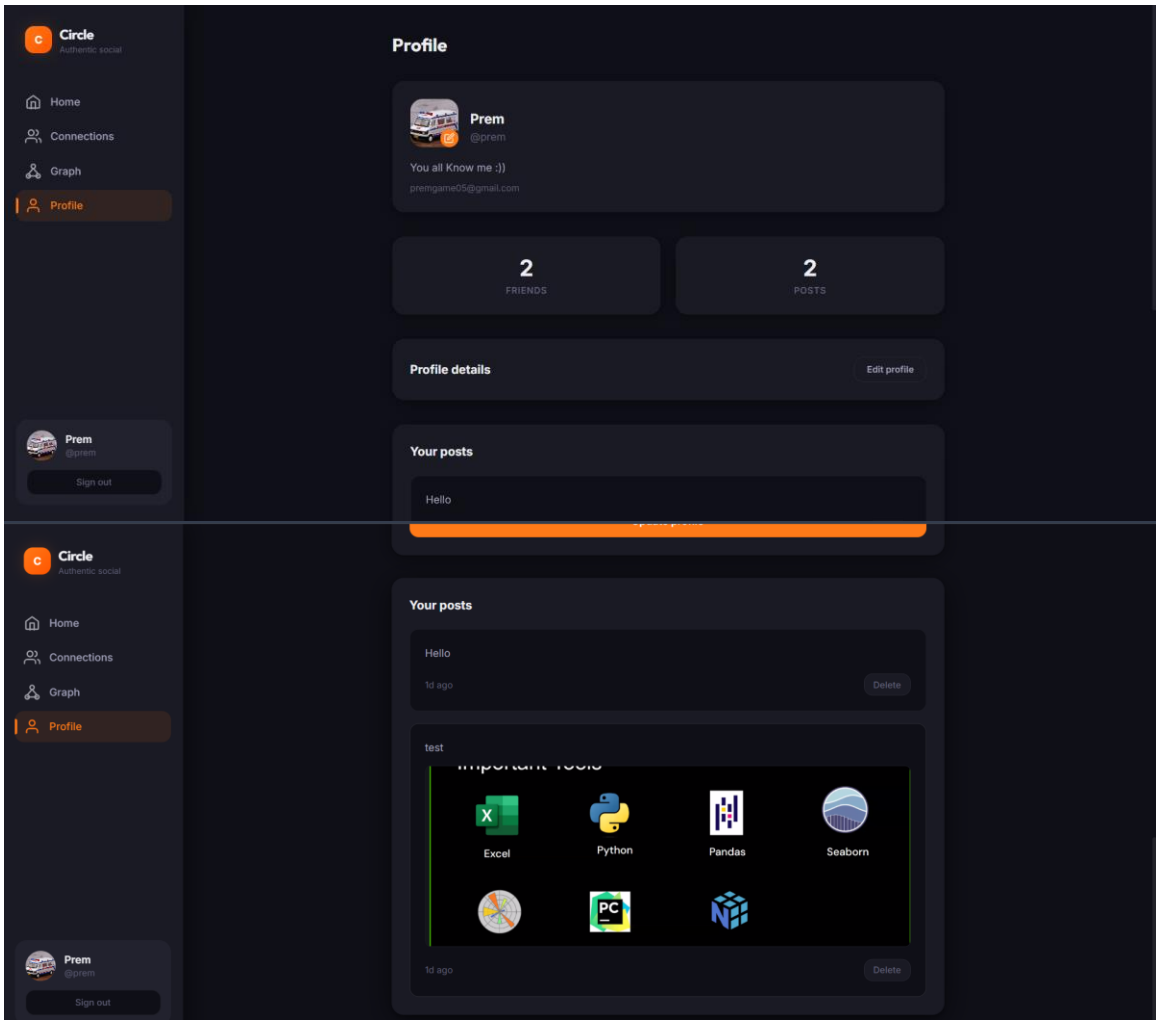
Connections Page



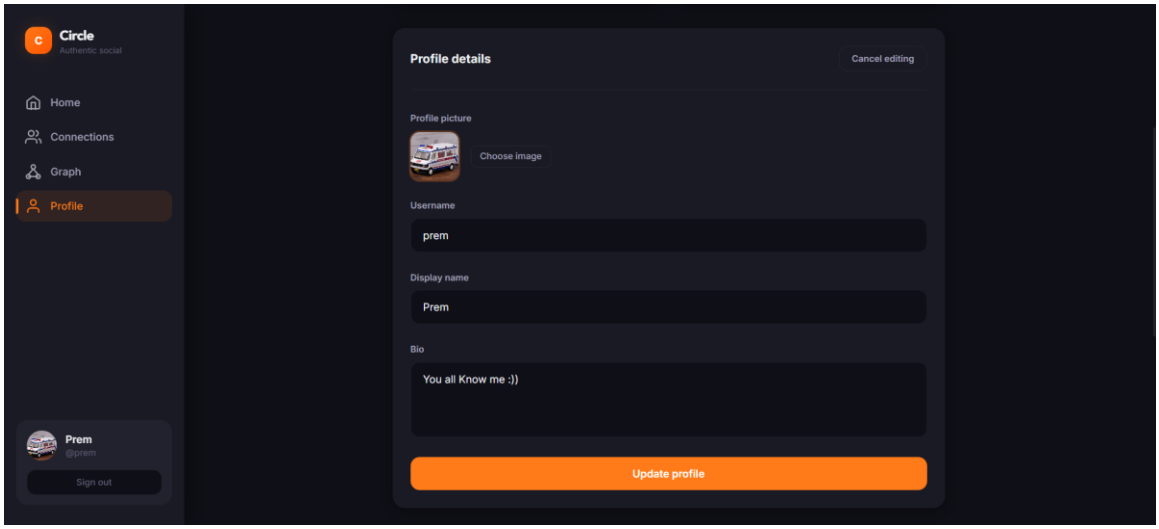
Friend Graph



Profile Page



Edit Profile



13. Learning Outcomes

- Developed a full-stack application using modern tools.

- Understood database design and relationships.

- Implemented secure access control using RLS.

- Built interactive UI with React and D3.

- Learned integration of frontend and backend systems.

14. References

https://supabase.com/docs

https://react.dev

https://tailwindcss.com

https://d3js.org